

Math 432/532: Introduction to Topology II

Solutions to HW #5

1. Let $\varphi : V \rightarrow U$ be a linear map. Prove that $V/\text{Ker}(\varphi)$ is isomorphic to $\text{Im}(\varphi)$.

Solution: Consider the map $\bar{\varphi} : V/\text{Ker}(\varphi) \rightarrow U$ given by the formula $\bar{\varphi}([v]) = \varphi(v)$ for all $v \in V$. It is clear that the image of $\bar{\varphi}$ is equal to the image of φ . If $\varphi(v) = 0$, then $v \in \text{Ker}(\varphi)$, so $[v] = 0$, thus $\bar{\varphi}$ is injective. The statement now follows from the fact that an injective linear map is an isomorphism onto its image.

2. Let K be the simplicial complex shown in Figure 6.4 of the book, except with the direction of the arrow on the right flipped, so that $|K|$ is homeomorphic to the Klein bottle. Compute the Betti numbers of K . (You may use the fact that the Klein bottle is connected and non-orientable.)

Solution: Since the Klein bottle is connected, $\beta_0 = 1$. Since it is not orientable, $\beta_2 = 0$. That means we only need to compute β_1 . We have $\alpha_0 = 9$, $\alpha_1 = 27$, and $\alpha_2 = 18$. Since $\beta_0 = 1$, the map ∂_1 has an image of dimension 8, and therefore a kernel of dimension 19. Since $\beta_2 = 0$, the map ∂_2 is injective, and therefore has an image of dimension 18. So $\beta_1 = 19 - 18 = 1$.

3. Let Γ be a tree, that is, a connected 1-dimensional simplicial complex with n vertices and $n - 1$ edges for some $n > 0$. Compute the Betti numbers of Γ .

Solution: Since Γ is connected, $\beta_0 = 1$. That means that ∂_1 must be injective, so $\beta_1 = 0$.

4. Let Γ be the complete graph on 4 vertices, meaning that it is a connected 1-dimensional simplicial complex with 4 vertices and 6 edges. Compute the Betti numbers of Γ .

Solution: Since Γ is connected, $\beta_0 = 1$. That means that ∂_1 has a 3-dimensional image, and therefore a 3-dimensional kernel, so $\beta_1 = 3$.

5. Let Δ_1 be the simplex in $\mathbb{R}^3$ with vertices $\{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. Let Δ_2 be the simplex in $\mathbb{R}^3$ with vertices $\{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, -1)\}$. Let K be the simplicial complex consisting of all proper faces of Δ_1 along with all proper faces of Δ_2 . Thus the realization $|K|$ is homeomorphic to two copies of S^2 , glued together along a copy of D^2 . Compute the Betti numbers of K .

Solution: This solution is modeled on the computation of the homology groups of the sphere (Example 5 from Section 8.3). Let $L = K \cup \{\Delta_1, \Delta_2\}$, and let M be the subcomplex of L consisting of simplices that do not contain the vertex $(1, 0, 0)$. Then $L = \text{Cone}(M, (1, 0, 0))$, and therefore we have $\beta_0(L) = 1$ and $\beta_i(L) = 0$ for all $i > 0$.

Observe that $C_i(K) = C_i(L)$ for all $i < 2$, so $\beta_0(K) = \beta_0(L) = 1$ and $\beta_1(K) = \beta_1(L) = 0$. We also have $C_i(K) = 0$ for all $i > 2$, so $\beta_i(K) = 0$ for all $i > 2$. That means that the only thing left to compute is $\beta_2(K)$.

We have $H_2(K) = Z_2(K) = Z_2(L) = B_2(L)$. This space is spanned by $\partial_3(\sigma_1)$ and $\partial_3(\sigma_2)$, where σ_1 and σ_2 are orientations of Δ_1 and Δ_2 , respectively. Since these classes are clearly not multiples of each other, we have $\beta_2(K) = 2$.